

Color Model

What is Color?

- It is an attribute of objects (like texture, shape, smoothness, etc.)
- It depends on:
 - 1) spectral characteristics of the light source(s) (e.g., sunlight)
 - 2) spectral properties of objects (reflectance)
 - 3) spectral characteristics of the sensors of the imaging device (e.g., the human eye or a digital camera)

Primary and Secondary Color

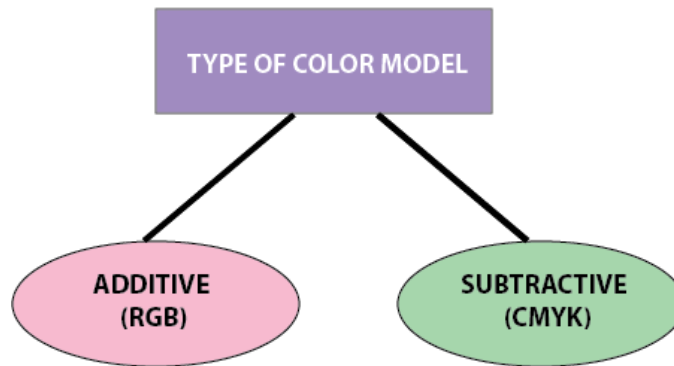
- Primary colors: red, green, and blue
- The primary colors can be added to produce the secondary colors of light, *magenta* (R+B), *cyan* (G+B), and *yellow* (R+G)

Color Model

- A color model is an abstract mathematical model describing the way colors can be represented as tuples of numbers, typically as three or four values or color components.
- A Color model is a hierarchical system in which we can create every color by using RGB (Red, Green, Blue) and CMYK (Cyan, Magenta, Yellow, Black) models. We can use different colors for various purposes.

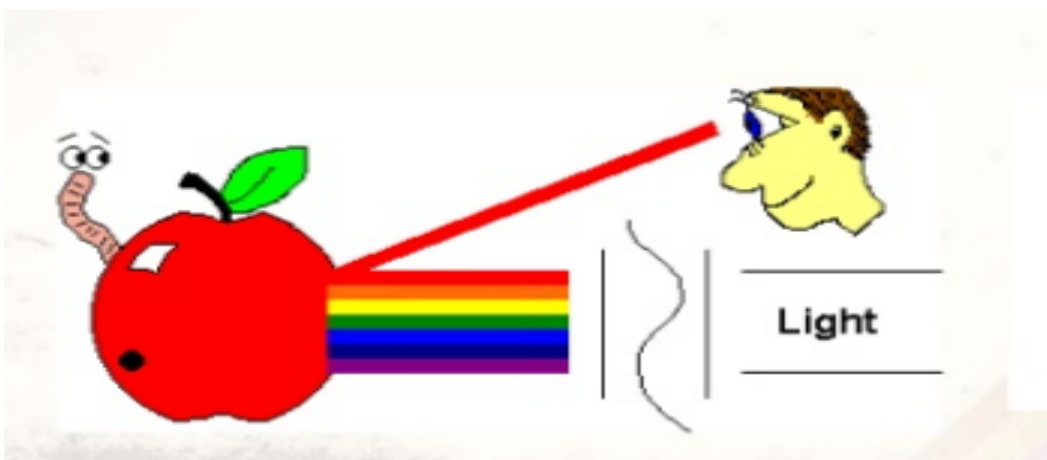
Different Color Models

The basic color model is divided into two parts-



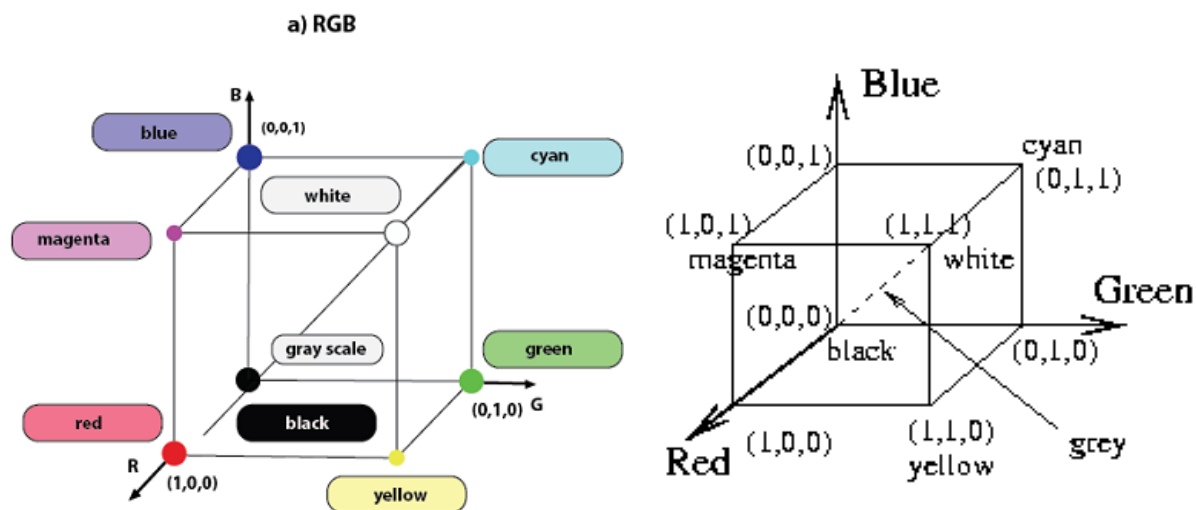
Properties of light

- When white light is incident on an opaque object, some frequencies are reflected and some are absorbed.
- The combination of frequencies present in the reflected light determines the color of the object that we see.



RGB Model

- In this model, the primary colors are red, green, and blue. It is an additive model, in which colors are produced by adding components, with white having all colors present and black being the absence of any color.
- This is the model used for active displays such as television and computer screens.
- The RGB model is usually represented by a unit cube with one corner located at the origin of a three-dimensional color coordinate system, the axes being labeled R, G, B, and having a range of values [0, 1]. The origin (0, 0, 0) is considered black and the diagonally opposite corner (1, 1, 1) is called white. The line joining black to white represents a gray scale and has equal components of R, G, B.



- Each color is represented in its primary color components **Red**, **Green** and **Blue**

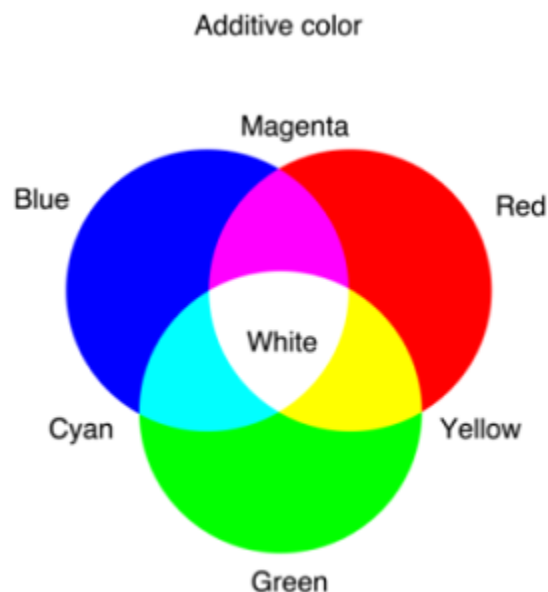
- This model is based on [Cartesian Coordinate System](#)

RGB color system

- Additive color model.
- For computer display.
- Uses light to display color.
- Colors result from transmitted light.
- Red + Green + Blue = White.

RGB color system

- This model is called additive, and the colors are called primary colors.
- The primary colors can be added to produce the secondary colors of light- Magenta(Red + Blue), Cyan(Green + Blue), and Yellow(Red + Green).
- The combination of Red, Green and Blue at full intensities makes white.

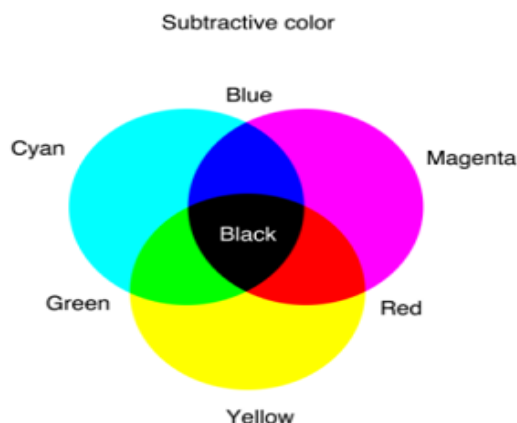


Importance of RGB color model

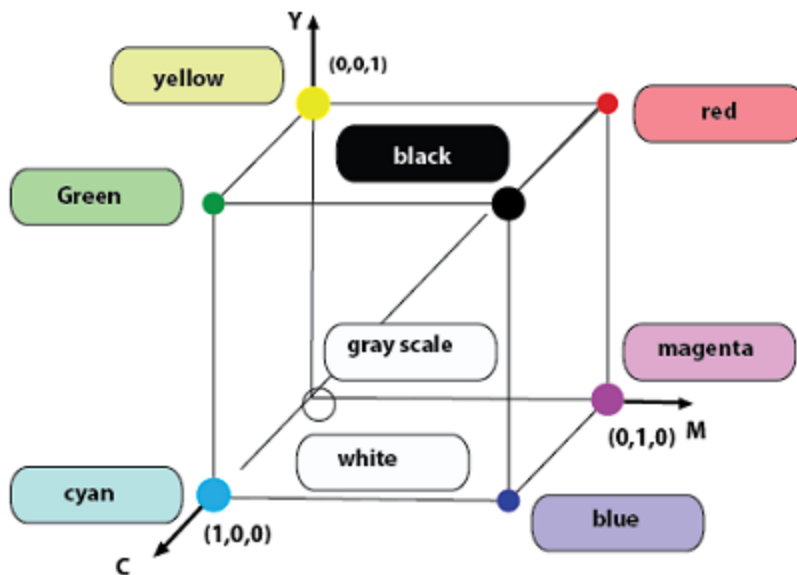
- The color model RGB is used in hardware applications like PC monitors, cameras and scanners.
- It is used for Web graphics, but it cannot be used for print production.
- It directly reflects the physical properties of “True-color” displays.
- It is used for—
 - Display of text images in electronic system, For example- computer, TV, camera.

Cmyk color model

- CMYK (subtractive color model) is the standard color model used in offset printing for full-color documents. Because such printing uses inks of these four basic colors, it is often called four-color printing.
- A greenish blue called Cyan.
- A blushed red called Magenta.
- A bright Yellow.
- The key color Black

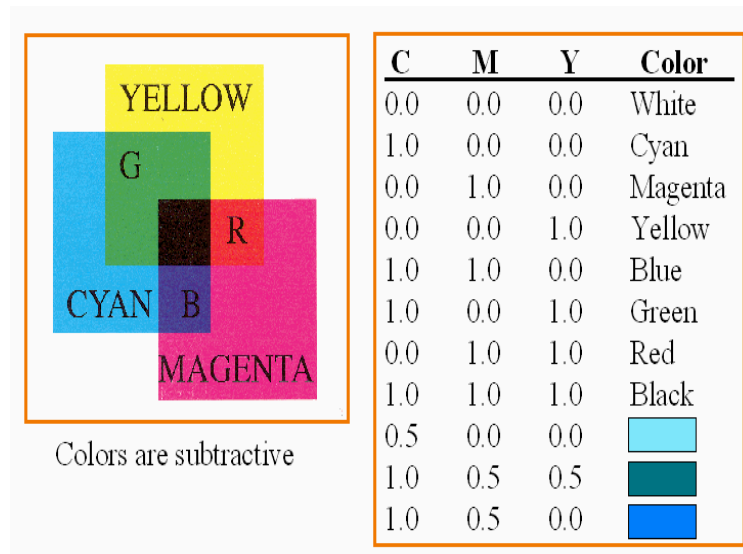


b) CMY



Cmyk color model system

- Subtractive color model.
- For printed material.
- Uses ink to display color.
- Colors result from reflected light.
- Cyan + Magenta + Yellow = Black.



Importance of cmyk color model

- The CMY color model is used in color printers.
- It is created by the subtractive mode
- Used in most commercial color printing (Books, Magazines etc.).